1.44 Matrices: Inverses & Applications

Worked Solutions

1)

A and B are inverse matrices, so $B = A^{-1}$ or $A = B^{-1}$.

2)

If A is singular, then $\det A = 0$, so A does not have an inverse matrix, as a step in finding the inverse matrix requires you to divide by $\det A$.

3 a)

$$\begin{vmatrix} 3 & 7 \\ 2 & 5 \end{vmatrix} = 3 \times 5 - 7 \times 2 = 1$$

$$\begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} 5 & -7 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 5 & -7 \\ -2 & 3 \end{pmatrix}$$

3 b)

$$\begin{vmatrix} 8 & 13 \\ -3 & -5 \end{vmatrix} = 8 \times (-5) - 13 \times (-3) = -1$$

$$\begin{pmatrix} 8 & 13 \\ -3 & -5 \end{pmatrix}^{-1} = \frac{1}{-1} \begin{pmatrix} -5 & -13 \\ 3 & 8 \end{pmatrix} = \begin{pmatrix} 5 & 13 \\ -3 & -8 \end{pmatrix}$$

3 c)

$$\begin{vmatrix} 9 & 1 \\ 28 & 3 \end{vmatrix} = 9 \times 3 - 1 \times 28 = -1$$

$$\begin{pmatrix} 9 & 1 \\ 28 & 3 \end{pmatrix}^{-1} = \frac{1}{-1} \begin{pmatrix} 3 & -1 \\ -28 & 9 \end{pmatrix} = \begin{pmatrix} -3 & 1 \\ 28 & -9 \end{pmatrix}$$

3 d)

$$\begin{vmatrix} 6 & 17 \\ 2 & 6 \end{vmatrix} = 6 \times 6 - 17 \times 2 = 2$$

$$\begin{pmatrix} 6 & 17 \\ 2 & 6 \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} 6 & -17 \\ -2 & 6 \end{pmatrix} = \begin{pmatrix} 3 & -\frac{17}{2} \\ -1 & 3 \end{pmatrix}$$

3 e)

$$\begin{vmatrix} -8 & -7 \\ 11 & 10 \end{vmatrix} = -8 \times 10 - (-7) \times 11 = -3$$

$$\begin{pmatrix} -8 & -7 \\ 11 & 10 \end{pmatrix}^{-1} = \frac{1}{-3} \begin{pmatrix} 10 & 7 \\ -11 & -8 \end{pmatrix} = \begin{pmatrix} -\frac{10}{3} & -\frac{7}{3} \\ \frac{11}{3} & \frac{8}{3} \end{pmatrix}$$

3 f)

$$\begin{vmatrix} 12 & 20 \\ 5 & 10 \end{vmatrix} = 12 \times 10 - 20 \times 5 = 20$$

$$\begin{pmatrix} 12 & 20 \\ 5 & 10 \end{pmatrix}^{-1} = \frac{1}{20} \begin{pmatrix} 10 & -20 \\ -5 & 12 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -1 \\ -\frac{1}{4} & \frac{3}{5} \end{pmatrix}$$

3 g)

$$\begin{vmatrix} -2 & 9 \\ 1 & 3 \end{vmatrix} = -2 \times 3 - 9 \times 1 = -15$$

$$\begin{pmatrix} -2 & 9 \\ 1 & 3 \end{pmatrix}^{-1} = \frac{1}{-15} \begin{pmatrix} 3 & -9 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{5} & \frac{3}{5} \\ \frac{1}{15} & \frac{2}{15} \end{pmatrix}$$

3 h)

$$\begin{vmatrix} 6 & -5 \\ -25 & 0 \end{vmatrix} = 6 \times 0 - (-5) \times (-25) = -125$$

$$\begin{pmatrix} 6 & -5 \\ -25 & 0 \end{pmatrix}^{-1} = \frac{1}{-125} \begin{pmatrix} 0 & 5 \\ 25 & 6 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{25} \\ -\frac{1}{5} & -\frac{6}{125} \end{pmatrix}$$

3 i)

$$\begin{vmatrix} -3 & 8 \\ -9 & 7 \end{vmatrix} = (-3) \times 7 - 8 \times (-9) = 51$$

$$\begin{pmatrix} -3 & 8 \\ -9 & 7 \end{pmatrix}^{-1} = \frac{1}{51} \begin{pmatrix} 7 & -8 \\ 9 & -3 \end{pmatrix} = \begin{pmatrix} \frac{7}{51} & -\frac{8}{51} \\ \frac{3}{17} & -\frac{1}{17} \end{pmatrix}$$

3 j)

$$\begin{vmatrix} 20 & -30 \\ -50 & -40 \end{vmatrix} = 20 \times (-40) - (-30) \times (-50) = -2300$$

$$\begin{pmatrix} 20 & -30 \\ -50 & -40 \end{pmatrix}^{-1} = \frac{1}{-2300} \begin{pmatrix} -40 & 30 \\ 50 & 20 \end{pmatrix} = \begin{pmatrix} \frac{2}{115} & -\frac{3}{230} \\ -\frac{1}{46} & -\frac{1}{115} \end{pmatrix}$$

4 a)

$$\det A = 0 \Rightarrow k \times 3 - 4 \times 2 = 0 \Rightarrow 3k - 8 = 0 \Rightarrow k = \frac{8}{3}.$$

4 b)

$$\begin{pmatrix} k & 4 \\ 2 & 3 \end{pmatrix}^{-1} = \frac{1}{3k-8}\begin{pmatrix} 3 & -4 \\ -2 & k \end{pmatrix} = \begin{pmatrix} \frac{3}{3k-8} & -\frac{4}{3k-8} \\ -\frac{2}{3k-8} & \frac{k}{3k-8} \end{pmatrix}.$$

5 a)

$$\det A = 0 \Rightarrow k \times k - 2 \times 2 = 0 \Rightarrow k^2 - 4 = 0 \Rightarrow k = \pm 2.$$

As k is a positive constant, $k = 2$.

5 b)

$$\begin{pmatrix} k & 2 \\ 2 & k \end{pmatrix}^{-1} = \frac{1}{k^2-4}\begin{pmatrix} k & -2 \\ -2 & k \end{pmatrix} = \begin{pmatrix} \frac{k}{k^2-4} & -\frac{2}{k^2-4} \\ -\frac{2}{k^2-4} & \frac{k}{k^2-4} \end{pmatrix}.$$

6 a)

$$\det A = 0 \Rightarrow (k+1)(k-1) - (-2\sqrt{2})(-2\sqrt{2}) = 0 \Rightarrow k^2 - 1 - 8 = 0$$
$$\Rightarrow k^2 - 9 = 0 \Rightarrow k = \pm 3.$$

As k is a negative constant, $k = -3$.

6 b)

$$\begin{pmatrix} k+1 & -2\sqrt{2} \\ -2\sqrt{2} & k-1 \end{pmatrix}^{-1} = \frac{1}{k^2-9}\begin{pmatrix} k-1 & 2\sqrt{2} \\ 2\sqrt{2} & k+1 \end{pmatrix} = \begin{pmatrix} \frac{k-1}{k^2-9} & \frac{2\sqrt{2}}{k^2-9} \\ \frac{2\sqrt{2}}{k^2-9} & \frac{k+1}{k^2-9} \end{pmatrix}.$$

7 a)

i) Enlargement, centred at the origin, scale factor 3.

Alternative Method: Stretch, parallel to the x-axis, scale factor 3, and a stretch, parallel to the y-axis, scale factor 3.

$$\begin{vmatrix} 3 & 0 \\ 0 & 3 \end{vmatrix} = 3 \times 3 - 0 \times 0 = 9$$

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}^{-1} = \frac{1}{9}\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$.

iii) Enlargement, centred at the origin, scale factor $\frac{1}{3}$.

Alternative Method: Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$, and a stretch, parallel to the y-axis, scale factor $\frac{1}{3}$.

7 b)

i) Reflection in the line $y = x$.

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0 \times 0 - 1 \times 1 = -1$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^{-1} = \frac{1}{-1}\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

iii) Reflection in the line $y = x$.

7 c)

i) Stretch, parallel to the x-axis, scale factor 8.

$$\begin{vmatrix} 8 & 0 \\ 0 & 1 \end{vmatrix} = 8 \times 1 - 0 \times 0 = 8$$

$$\begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{8}\begin{pmatrix} 1 & 0 \\ 0 & 8 \end{pmatrix} = \begin{pmatrix} \frac{1}{8} & 0 \\ 0 & 1 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} \frac{1}{8} & 0 \\ 0 & 1 \end{pmatrix}$.

iii) Stretch, parallel to the x-axis, scale factor $\frac{1}{8}$.

7 d)

i) Rotation, anticlockwise about the origin, by 90°. Alternative Method: Rotation, clockwise about the origin, by 270°.

$$\begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} = 0 \times 0 - (-1) \times 1 = 1$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

iii) Rotation, anticlockwise about the origin, by 270°. Alternative Method: Rotation, clockwise about the origin, by 90°.

7 e)

i) Reflection in the x-axis.

$$\begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = 1 \times (-1) - 0 \times 0 = -1$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^{-1} = \frac{1}{-1}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

iii) Reflection in the x-axis.

7 f)

i) Rotation, anticlockwise about the origin, by 180°.

$$\begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix} = (-1) \times (-1) - 0 \times 0 = 1$$

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

iii) Rotation, anticlockwise about the origin, by 180°.

7 g)

i) Rotation, anticlockwise about the origin, by 270°. Alternative Method: Rotation, clockwise about the origin, by 90°.

$$\begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} = 0 \times 0 - 1 \times (-1) = 1$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

iii) Rotation, anticlockwise about the origin, by 90°. Alternative Method: Rotation, clockwise about the origin, by 270°.

7 h)

i) Reflection in the y-axis.

$$\begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} = (-1) \times 1 - 0 \times 0 = -1$$

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{-1} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

iii) Reflection in the y-axis.

7 i)

i) Rotation, anticlockwise about the origin, by 45°. Alternative Method: Rotation, clockwise about the origin, by 315°.

$$\begin{vmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{vmatrix} = \frac{1}{2} + \frac{1}{2} = 1$$

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}^{-1} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$.

iii) Rotation, anticlockwise about the origin, by 315°. Alternative Method: Rotation, clockwise about the origin, by 45°.

7 j)

i) Rotation, anticlockwise about the origin, by 225° (or clockwise by 135°).

$$\begin{vmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{vmatrix} = \frac{1}{2} + \frac{1}{2} = 1$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}^{-1} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$.

iii) Rotation, anticlockwise about the origin, by 135°. Alternative Method: Rotation, clockwise about the origin, by 225°.

7 k)

i) Rotation, anticlockwise about the origin, by 30°. Alternative Method: Rotation, clockwise about the origin, by 330°.

$$\begin{vmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{vmatrix} = \frac{3}{4} + \frac{1}{4} = 1$$

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}^{-1} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$.

iii) Rotation, anticlockwise about the origin, by 330°. Alternative Method: Rotation, clockwise about the origin, by 30°.

7 l)

i) Rotation, anticlockwise about the origin, by 300° (or clockwise by 60°).

$$\begin{vmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{4} + \frac{3}{4} = 1$$

$$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$.

iii) Rotation, anticlockwise about the origin, by 60°. Alternative Method: Rotation, clockwise about the origin, by 300°.

8 a)

i) Shear, with the x-axis fixed, with $(0, 1)$ mapped to $(4, 1)$.

$$\begin{vmatrix} 1 & 4 \\ 0 & 1 \end{vmatrix} = 1 \times 1 - 4 \times 0 = 1$$

$$\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}$.

iii) Shear, with the x-axis fixed, with $(0, 1)$ mapped to $(-4, 1)$.

8 b)

i) Shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, 6)$.

$$\begin{vmatrix} 1 & 0 \\ 6 & 1 \end{vmatrix} = 1 \times 1 - 0 \times 6 = 1$$

$$\begin{pmatrix} 1 & 0 \\ 6 & 1 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} 1 & 0 \\ -6 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -6 & 1 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} 1 & 0 \\ -6 & 1 \end{pmatrix}$.

iii) Shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, -6)$.

8 c)

i) Shear, with the x-axis fixed, with $(0, 1)$ mapped to $(-2, 1)$.

$$\begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} = 1 \times 1 - (-2) \times 0 = 1$$

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

iii) Shear, with the x-axis fixed, with $(0, 1)$ mapped to $(2, 1)$.

8 d)

i) Shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, -\frac{2}{5})$.

$$\begin{vmatrix} 1 & 0 \\ -\frac{2}{5} & 1 \end{vmatrix} = 1 \times 1 - 0 \times (-\frac{2}{5}) = 1$$

$$\begin{pmatrix} 1 & 0 \\ -\frac{2}{5} & 1 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} 1 & 0 \\ \frac{2}{5} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{2}{5} & 1 \end{pmatrix}$$

ii) The inverse matrix is $\begin{pmatrix} 1 & 0 \\ \frac{2}{5} & 1 \end{pmatrix}$.

iii) Shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, \frac{2}{5})$.

9 a)

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

9 b)

$$B = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

9 c)

$$AB = I \Rightarrow \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \cos^2\theta + \sin^2\theta & \sin\theta\cos\theta - \sin\theta\cos\theta \\ \sin\theta\cos\theta - \sin\theta\cos\theta & \sin^2\theta + \cos^2\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \sin^2\theta + \cos^2\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\therefore \sin^2\theta + \cos^2\theta \equiv 1.$$

10 a)

$$\det A = 2 \times 5 - 3 \times 3 = 1$$

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}.$$

10 b)

Let the coordinates of P be (a, b).

$$A \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \Rightarrow A^{-1}A \begin{pmatrix} a \\ b \end{pmatrix} = A^{-1} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = A^{-1} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}.$$

So P has coordinates $(-4, 3)$.

11 a)

$$\det A = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right) \times \frac{1}{\sqrt{2}} = 1$$

$$A^{-1} = \frac{1}{1} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

11 b)

Let the coordinates of P be (a, b).

$$A \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \Rightarrow A^{-1}A \begin{pmatrix} a \\ b \end{pmatrix} = A^{-1} \begin{pmatrix} -3 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = A^{-1} \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} -3 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sqrt{2} \\ 2\sqrt{2} \end{pmatrix}.$$

So P has coordinates $(-\sqrt{2}, 2\sqrt{2})$.

12)

$$AB = C \Rightarrow A^{-1}AB = A^{-1}C \Rightarrow IB = A^{-1}C \Rightarrow B = A^{-1}C$$

$$\det A = 3 \times 1 - 2 \times 1 = 1$$

$$A^{-1} = \frac{1}{1}\begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$$

$$\therefore B = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 3 \end{pmatrix}.$$

13)

$$AB = C \Rightarrow ABB^{-1} = CB^{-1} \Rightarrow AI = CB^{-1} \Rightarrow A = CB^{-1}$$

$$\det B = 4 \times 5 - 3 \times 6 = 2$$

$$B^{-1} = \frac{1}{2}\begin{pmatrix} 5 & -3 \\ -6 & 4 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} & -\frac{3}{2} \\ -3 & 2 \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} 6 & 2 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} \frac{5}{2} & -\frac{3}{2} \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 9 & -5 \\ 7 & -4 \end{pmatrix}.$$

14)

$$AB = \begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix} \Rightarrow \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} B = \begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} B = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix}$$

$$\Rightarrow IB = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix} \Rightarrow B = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix}$$

$$\begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 3 \times 2 - 1 \times 4 = 2$$

$$\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1} = \frac{1}{2}\begin{pmatrix} 2 & -1 \\ -4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{pmatrix}$$

$$\therefore B = \begin{pmatrix} 1 & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{pmatrix}\begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 1 & \frac{1}{2} \\ -7 & -4 & \frac{3}{2} \end{pmatrix}.$$

15)

$$AB = \begin{pmatrix} 34 & -1 \\ -24 & 11 \\ 16 & 1 \end{pmatrix} \Rightarrow A\begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 34 & -1 \\ -24 & 11 \\ 16 & 1 \end{pmatrix}$$

$$\Rightarrow A\begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix}\begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 34 & -1 \\ -24 & 11 \\ 16 & 1 \end{pmatrix}\begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix}^{-1}$$

$$\Rightarrow AI = \begin{pmatrix} 34 & -1 \\ -24 & 11 \\ 16 & 1 \end{pmatrix} \begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix}^{-1} \Rightarrow A = \begin{pmatrix} 34 & -1 \\ -24 & 11 \\ 16 & 1 \end{pmatrix} \begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix}^{-1}$$

$$\begin{vmatrix} 9 & -1 \\ -2 & 3 \end{vmatrix} = 9 \times 3 - (-1) \times (-2) = 25$$

$$\begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix}^{-1} = \frac{1}{25} \begin{pmatrix} 3 & 1 \\ 2 & 9 \end{pmatrix} = \begin{pmatrix} \frac{3}{25} & \frac{1}{25} \\ \frac{2}{25} & \frac{9}{25} \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} 34 & -1 \\ -24 & 11 \\ 16 & 1 \end{pmatrix} \begin{pmatrix} \frac{3}{25} & \frac{1}{25} \\ \frac{2}{25} & \frac{9}{25} \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ -2 & 3 \\ 2 & 1 \end{pmatrix}.$$

16)

$$AB = \begin{pmatrix} 9 & 33 \\ 15 & 39 \\ 11 & 26 \\ 21 & 54 \end{pmatrix} \Rightarrow A \begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} 9 & 33 \\ 15 & 39 \\ 11 & 26 \\ 21 & 54 \end{pmatrix}$$

$$\Rightarrow A \begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix}^{-1} = \begin{pmatrix} 9 & 33 \\ 15 & 39 \\ 11 & 26 \\ 21 & 54 \end{pmatrix} \begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix}^{-1}$$

$$\Rightarrow AI = \begin{pmatrix} 9 & 33 \\ 15 & 39 \\ 11 & 26 \\ 21 & 54 \end{pmatrix} \begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix}^{-1} \Rightarrow A = \begin{pmatrix} 9 & 33 \\ 15 & 39 \\ 11 & 26 \\ 21 & 54 \end{pmatrix} \begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix}^{-1}$$

$$\begin{vmatrix} 5 & 3 \\ 2 & 6 \end{vmatrix} = 5 \times 6 - 3 \times 2 = 24$$

$$\begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix}^{-1} = \frac{1}{24} \begin{pmatrix} 6 & -3 \\ -2 & 5 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & -\frac{1}{8} \\ -\frac{1}{12} & \frac{5}{24} \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} 9 & 33 \\ 15 & 39 \\ 11 & 26 \\ 21 & 54 \end{pmatrix} \begin{pmatrix} \frac{1}{4} & -\frac{1}{8} \\ -\frac{1}{12} & \frac{5}{24} \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 2 & 4 \\ 1 & 2 \\ 3 & 8 \end{pmatrix}.$$

17 a)

$$\det A = 2 \times 8 - 4 \times (-9) = 52$$

$$A^{-1} = \frac{1}{52} \begin{pmatrix} 8 & -4 \\ 9 & 2 \end{pmatrix} = \begin{pmatrix} \frac{2}{13} & -\frac{1}{13} \\ \frac{9}{52} & \frac{1}{26} \end{pmatrix}.$$

17 b)

$$AB = \begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix} \Rightarrow A^{-1}AB = A^{-1}\begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix} \Rightarrow IB = A^{-1}\begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix}$$

$$\Rightarrow B = A^{-1}\begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix} = \begin{pmatrix} \frac{2}{13} & -\frac{1}{13} \\ \frac{9}{52} & \frac{1}{26} \end{pmatrix}\begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$$

$$C = 3A - 4B = 3\begin{pmatrix} 2 & 4 \\ -9 & 8 \end{pmatrix} - 4\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 12 \\ -27 & 24 \end{pmatrix} - \begin{pmatrix} 12 & -20 \\ -4 & 8 \end{pmatrix} = \begin{pmatrix} -6 & 32 \\ -23 & 16 \end{pmatrix}.$$

18)

$$AA = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Hence the matrix A is self-inverse.

19)

Let triangle T have vertices at (a, b), (c, d) and (e, f), so that when T is mapped to T', the vertices (a, b), (c, d) and (e, f) are mapped to $(2, 3)$, $(4, -5)$ and $(-1, -1)$ respectively.

$$\begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}\begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}\begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix}$$

$$\begin{vmatrix} 2 & 1 \\ 13 & 7 \end{vmatrix} = 2 \times 7 - 1 \times 13 = 1$$

$$\begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 7 & -1 \\ -13 & 2 \end{pmatrix} = \begin{pmatrix} 7 & -1 \\ -13 & 2 \end{pmatrix}$$

$$\therefore \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 7 & -1 \\ -13 & 2 \end{pmatrix}\begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix} = \begin{pmatrix} 11 & 33 & -6 \\ -20 & -62 & 11 \end{pmatrix}.$$

So the vertices of T are $(11, -20)$, $(33, -62)$ and $(-6, 11)$.

20)

$$AB = C \Rightarrow A^{-1}AB = A^{-1}C \Rightarrow IB = A^{-1}C \Rightarrow B = A^{-1}C$$

$$\det A = p \times p - 4q \times 2q = p^2 - 8q^2$$

$$A^{-1} = \frac{1}{p^2 - 8q^2}\begin{pmatrix} p & -2q \\ -4q & p \end{pmatrix}$$

$$\therefore B = \frac{1}{p^2-8q^2}\begin{pmatrix} p & -2q \\ -4q & p \end{pmatrix}\begin{pmatrix} 3p & q \\ 5q & 2p \end{pmatrix} = \frac{1}{p^2-8q^2}\begin{pmatrix} 3p^2-10q^2 & pq-4pq \\ -12pq+5pq & -4q^2+2p^2 \end{pmatrix}$$
$$= \frac{1}{p^2-8q^2}\begin{pmatrix} 3p^2-10q^2 & -3pq \\ -7pq & 2p^2-4q^2 \end{pmatrix} = \begin{pmatrix} \frac{3p^2-10q^2}{p^2-8q^2} & -\frac{3pq}{p^2-8q^2} \\ -\frac{7pq}{p^2-8q^2} & \frac{2p^2-4q^2}{p^2-8q^2} \end{pmatrix}.$$

21)

$$AB = C \Rightarrow \det(AB) = \det C \Rightarrow \det A \times \det B = \det C$$
$$\Rightarrow \det B = \frac{\det C}{\det A} = \frac{5p \times p - (-2q) \times (-4q)}{3p \times (-p) - 4q \times 2q} = \frac{5p^2-8q^2}{-3p^2-8q^2}.$$

Alternative Method:

$$AB = C \Rightarrow A^{-1}AB = A^{-1}C \Rightarrow IB = A^{-1}C \Rightarrow B = A^{-1}C$$
$$\det A = 3p \times (-p) - 4q \times 2q = -3p^2 - 8q^2$$
$$A^{-1} = \frac{1}{-3p^2-8q^2}\begin{pmatrix} -p & -4q \\ -2q & 3p \end{pmatrix}$$
$$\therefore B = \frac{1}{-3p^2-8q^2}\begin{pmatrix} -p & -4q \\ -2q & 3p \end{pmatrix}\begin{pmatrix} 5p & -2q \\ -4q & p \end{pmatrix} = \frac{1}{-3p^2-8q^2}\begin{pmatrix} -5p^2+16q^2 & 2pq-4pq \\ -10pq-12pq & 4q^2+3p^2 \end{pmatrix}$$
$$= \frac{1}{-3p^2-8q^2}\begin{pmatrix} -5p^2+16q^2 & -2pq \\ -22pq & 3p^2+4q^2 \end{pmatrix}$$
$$\det B = \left(\frac{1}{-3p^2-8q^2}\right)^2 \begin{vmatrix} -5p^2+16q^2 & -2pq \\ -22pq & 3p^2+4q^2 \end{vmatrix}$$
$$= \left(\frac{1}{-3p^2-8q^2}\right)^2 ((-5p^2+16q^2)(3p^2+4q^2) - (-2pq)(-22pq))$$
$$= \left(\frac{1}{-3p^2-8q^2}\right)^2 (-15p^4 - 20p^2q^2 + 48p^2q^2 + 64q^4 - 44p^2q^2)$$
$$= \frac{-15p^4-16p^2q^2+64q^4}{(-3p^2-8q^2)^2} = \frac{(-3p^2-8q^2)(5p^2-8q^2)}{(-3p^2-8q^2)^2} = \frac{5p^2-8q^2}{-3p^2-8q^2}.$$

22)

$$\begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6k \\ -3 \end{pmatrix} \Rightarrow \begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix}^{-1}\begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix}^{-1}\begin{pmatrix} 6k \\ -3 \end{pmatrix}$$
$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix}^{-1}\begin{pmatrix} 6k \\ -3 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix}^{-1}\begin{pmatrix} 6k \\ -3 \end{pmatrix}$$
$$\begin{vmatrix} 3k & -5 \\ -k & 7 \end{vmatrix} = 3k \times 7 - (-5) \times (-k) = 21k - 5k = 16k$$
$$\begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix}^{-1} = \frac{1}{16k}\begin{pmatrix} 7 & 5 \\ k & 3k \end{pmatrix}$$
$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{16k}\begin{pmatrix} 7 & 5 \\ k & 3k \end{pmatrix}\begin{pmatrix} 6k \\ -3 \end{pmatrix} = \frac{1}{16k}\begin{pmatrix} 42k-15 \\ 6k^2-9k \end{pmatrix} = \begin{pmatrix} \frac{42k-15}{16k} \\ \frac{6k-9}{16} \end{pmatrix}$$
$$\therefore x = \frac{42k-15}{16k} \text{ and } y = \frac{6k-9}{16}.$$

23)

$$2A^{-1} + A = 7I \Rightarrow 2\begin{pmatrix} 3 & k \\ -2 & 4 \end{pmatrix}^{-1} + \begin{pmatrix} 3 & k \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$$

$$\Rightarrow 2 \times \frac{1}{12+2k}\begin{pmatrix} 4 & -k \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 3 & k \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$$

$$\Rightarrow \frac{1}{6+k}\begin{pmatrix} 4 & -k \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 3 & k \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4 & -k \\ 2 & 3 \end{pmatrix} + (6+k)\begin{pmatrix} 3 & k \\ -2 & 4 \end{pmatrix} = (6+k)\begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4 & -k \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 18+3k & 6k+k^2 \\ -12-2k & 24+4k \end{pmatrix} = \begin{pmatrix} 42+7k & 0 \\ 0 & 42+7k \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 22+3k & 5k+k^2 \\ -10-2k & 27+4k \end{pmatrix} = \begin{pmatrix} 42+7k & 0 \\ 0 & 42+7k \end{pmatrix}$$

$$22 + 3k = 42 + 7k \Rightarrow -20 = 4k \Rightarrow k = -5$$

$$5k + k^2 = 0 \Rightarrow k(5+k) = 0 \Rightarrow k = 0 \text{ or } k = -5$$

$$-10 - 2k = 0 \Rightarrow k = -5$$

$$27 + 4k = 42 + 7k \Rightarrow -15 = 3k \Rightarrow k = -5$$

$$\therefore k = -5.$$

24 a)

$$\det A = 5 \times 3 - 7 \times 2 = 1 \neq 0$$

$$\det B = 4 \times 6 - 3 \times 7 = 3 \neq 0$$

Therefore A and B are non-singular.

24 b)

$$AB = \begin{pmatrix} 5 & 7 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 4 & 3 \\ 7 & 6 \end{pmatrix} = \begin{pmatrix} 69 & 57 \\ 29 & 24 \end{pmatrix}$$

$$\det(AB) = 69 \times 24 - 57 \times 29 = 3$$

$$(AB)^{-1} = \frac{1}{3}\begin{pmatrix} 24 & -57 \\ -29 & 69 \end{pmatrix} = \begin{pmatrix} 8 & -19 \\ -\frac{29}{3} & 23 \end{pmatrix}$$

$$B^{-1} = \frac{1}{3}\begin{pmatrix} 6 & -3 \\ -7 & 4 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -\frac{7}{3} & \frac{4}{3} \end{pmatrix}$$

$$A^{-1} = \frac{1}{1}\begin{pmatrix} 3 & -7 \\ -2 & 5 \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ -2 & 5 \end{pmatrix}$$

$$B^{-1}A^{-1} = \begin{pmatrix} 2 & -1 \\ -\frac{7}{3} & \frac{4}{3} \end{pmatrix}\begin{pmatrix} 3 & -7 \\ -2 & 5 \end{pmatrix} = \begin{pmatrix} 8 & -19 \\ -\frac{29}{3} & 23 \end{pmatrix}$$

$$\therefore (AB)^{-1} = B^{-1}A^{-1}.$$

25 a)

$$\det A = 8 \times 2 - 2 \times 7 = 2$$
$$A^{-1} = \frac{1}{2}\begin{pmatrix} 2 & -2 \\ -7 & 8 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -\frac{7}{2} & 4 \end{pmatrix}.$$

25 b)

$$\det B = (-1) \times 2 - (-1) \times 3 = 1$$
$$B^{-1} = \frac{1}{1}\begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix}.$$

25 c)

$$(AB)^{-1} = B^{-1}A^{-1} = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -\frac{7}{2} & 4 \end{pmatrix} = \begin{pmatrix} -\frac{3}{2} & 2 \\ \frac{1}{2} & -1 \end{pmatrix}.$$

26 a)

$$\det A = 3 \times 1 - 2 \times 2 = -1$$
$$A^{-1} = \frac{1}{-1}\begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 2 & -3 \end{pmatrix}.$$

26 b)

$$\det B = 5 \times 2 - 3 \times 3 = 1$$
$$B^{-1} = \frac{1}{1}\begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix} = \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}.$$

26 c)

$$\det C = 6 \times 3 - 7 \times 3 = -3$$
$$C^{-1} = \frac{1}{-3}\begin{pmatrix} 3 & -7 \\ -3 & 6 \end{pmatrix} = \begin{pmatrix} -1 & \frac{7}{3} \\ 1 & -2 \end{pmatrix}.$$

26 d)

$$(ABC)^{-1} = C^{-1}B^{-1}A^{-1} = \begin{pmatrix} -1 & \frac{7}{3} \\ 1 & -2 \end{pmatrix}\begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} -1 & 2 \\ 2 & -3 \end{pmatrix} = \begin{pmatrix} \frac{115}{3} & -62 \\ -34 & 55 \end{pmatrix}.$$

27 a)

$$\begin{pmatrix} k & -4 \\ 2 & 3 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k^2 - 8 & -4k - 12 \\ 2k + 6 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$k^2 - 8 = 1 \Rightarrow k^2 = 9 \Rightarrow k = \pm 3$$

$$2k + 6 = 0 \Rightarrow k = -3$$

$$-4k - 12 = 0 \Rightarrow k = -3$$

The matrix is self-inverse when $k = -3$.

27 b)

$$\begin{pmatrix} k & -35 \\ 1 & -k \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k^2 - 35 & 0 \\ 0 & k^2 - 35 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$k^2 - 35 = 1 \Rightarrow k^2 = 36 \Rightarrow k = \pm 6.$$

The matrix is self-inverse when $k = 6$ or $k = -6$.

27 c)

$$\begin{pmatrix} -3 & k \\ -2k & 3 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} -2k^2 + 9 & 0 \\ 0 & -2k^2 + 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$-2k^2 + 9 = 1 \Rightarrow k^2 = 4 \Rightarrow k = \pm 2.$$

The matrix is self-inverse when $k = 2$ or $k = -2$.

27 d)

$$\begin{pmatrix} k+1 & 4 \\ -k & -3 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k^2 - 2k + 1 & 4k - 8 \\ -k^2 + 2k & -4k + 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$k^2 - 2k + 1 = 1 \Rightarrow k^2 - 2k = 0 \Rightarrow k(k-2) = 0 \Rightarrow k = 0 \text{ or } k = 2$$

$$-k^2 + 2k = 0 \Rightarrow k^2 - 2k = 0 \Rightarrow k(k-2) = 0 \Rightarrow k = 0 \text{ or } k = 2$$

$$4k - 8 = 0 \Rightarrow k = 2$$

$$-4k + 9 = 1 \Rightarrow k = 2.$$

The matrix is self-inverse when $k = 2$.

27 e)

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & 2k - 1 \\ -4k & \frac{\sqrt{2}}{2} \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{-16k^2+8k+1}{2} & 0 \\ 0 & \frac{-16k^2+8k+1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\frac{-16k^2 + 8k + 1}{2} = 1 \Rightarrow -16k^2 + 8k + 1 = 2 \Rightarrow -16k^2 + 8k - 1 = 0$$

$$\Rightarrow 16k^2 - 8k + 1 = 0 \Rightarrow (4k-1)^2 = 0 \Rightarrow k = \frac{1}{4}.$$

The matrix is self-inverse when $k = \frac{1}{4}$.

28)

If A is self-inverse then $A^{-1} = A$.

Pre-multiply by A:

$$A \times A^{-1} = A \times A \Rightarrow I = A^2.$$

29 a)

$$AB = kI \Rightarrow \begin{pmatrix} a & 6 \\ b & c \end{pmatrix}\begin{pmatrix} -3 & 2 \\ -4 & 6 \end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \Rightarrow \begin{pmatrix} -3a-24 & 2a+36 \\ -3b-4c & 2b+6c \end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$$

$$2a + 36 = 0 \Rightarrow a = -18$$

$$-3a - 24 = k \Rightarrow -3 \times (-18) - 24 = k \Rightarrow k = 30$$

$$-3b - 4c = 0 \quad (1) \qquad 2b + 6c = 30 \quad (2)$$

Therefore $b = -12$ and $c = 9$.

$$\therefore a = -18,\ b = -12,\ c = 9 \text{ and } k = 30.$$

29 b)

$$AB = 30I \Rightarrow A^{-1}AB = A^{-1} \times 30I \Rightarrow IB = 30A^{-1} \Rightarrow A^{-1} = \frac{1}{30}B$$

$$\therefore A^{-1} = \frac{1}{30}\begin{pmatrix} -3 & 2 \\ -4 & 6 \end{pmatrix} = \begin{pmatrix} -\frac{1}{10} & \frac{1}{15} \\ -\frac{2}{15} & \frac{1}{5} \end{pmatrix}.$$

30 a)

$$(BA)^{-1} = A^{-1}B^{-1} = \begin{pmatrix} \frac{5}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}$$

$$\begin{vmatrix} \frac{5}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{vmatrix} = \frac{5}{6} \times \frac{2}{3} - \left(-\frac{1}{3}\right) \times \left(-\frac{2}{3}\right) = \frac{1}{3}$$

$$\begin{pmatrix} \frac{5}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}^{-1} = \frac{1}{\frac{1}{3}}\begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{5}{6} \end{pmatrix} = 3\begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{5}{6} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix}$$

$$\therefore BA = \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix}.$$

30 b)

$$BA = \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix} \Rightarrow B^{-1}BA = B^{-1}\begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix} \Rightarrow IA = B^{-1}\begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix} \Rightarrow A = B^{-1}\begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix}$$

$$\det B = 1 \times \frac{3}{2} - 0 \times 1 = \frac{3}{2}$$

$$B^{-1} = \frac{1}{\frac{3}{2}}\begin{pmatrix} \frac{3}{2} & 0 \\ -1 & 1 \end{pmatrix} = \frac{2}{3}\begin{pmatrix} \frac{3}{2} & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} 1 & 0 \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

31 a)

$$\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{5}.$$

31 b)

$$\det(B^{-1}) = \frac{1}{\det B} = \frac{1}{\frac{2}{3}} = \frac{3}{2}.$$

31 c)

$$\det((AB)^{-1}) = \det(B^{-1}A^{-1}) = \det(B^{-1}) \times \det(A^{-1}) = \frac{3}{2} \times \frac{1}{5} = \frac{3}{10}.$$

32 a)

$$\det(A^{-1}) = 4 \Rightarrow \det A = \frac{1}{4}$$
$$\det(A^4) = (\det A)^4 = \left(\frac{1}{4}\right)^4 = \frac{1}{256}.$$

32 b)

$$\det(AB) = \det A \times \det B = \frac{1}{4} \times (-2) = -\frac{1}{2}.$$

33)

$$\det(A^{-1}) = -\frac{1}{5} \Rightarrow \det A = -5$$
$$\det(AB) = 40 \Rightarrow \det A \times \det B = 40 \Rightarrow -5 \times \det B = 40 \Rightarrow \det B = -8$$
$$\det(B^5) = (\det B)^5 = (-8)^5 = -32768.$$

34 a)

$$AB = \begin{pmatrix} 10 & -6 & 2 \\ 3 & -2 & 1 \\ -4 & 4 & -2 \end{pmatrix} \begin{pmatrix} 0 & 2 & 1 \\ -1 & 6 & 2 \\ -2 & 8 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

34 b)

$$AB = 2I \Rightarrow A^{-1}AB = A^{-1} \times 2I \Rightarrow IB = 2A^{-1} \Rightarrow \frac{1}{2}B = A^{-1}$$
$$\therefore A^{-1} = \frac{1}{2} \begin{pmatrix} 0 & 2 & 1 \\ -1 & 6 & 2 \\ -2 & 8 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & \frac{1}{2} \\ -\frac{1}{2} & 3 & 1 \\ -1 & 4 & \frac{1}{2} \end{pmatrix}.$$

35 a)

$$AB = kI \Rightarrow \begin{pmatrix} p & 0 & q \\ 2 & 1 & 4 \\ -2 & 0 & r \end{pmatrix} \begin{pmatrix} 0 & 0 & -2 \\ 4 & 4 & \frac{1}{3} \\ -1 & 0 & \frac{3}{4} \end{pmatrix} = \begin{pmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -q & 0 & \frac{3q}{4} - 2p \\ 0 & 4 & 0 \\ -r & 0 & \frac{3r}{4} + 4 \end{pmatrix} = \begin{pmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{pmatrix}$$

$k = 4 \qquad -q = k \Rightarrow -q = 4 \Rightarrow q = -4 \qquad -r = 0 \Rightarrow r = 0$

$$\frac{3q}{4} - 2p = 0 \Rightarrow \frac{3(-4)}{4} - 2p = 0 \Rightarrow p = -\frac{3}{2}.$$

35 b)

$$AB = 4I \Rightarrow A^{-1}AB = A^{-1} \times 4I \Rightarrow IB = 4A^{-1} \Rightarrow A^{-1} = \frac{1}{4}B$$

$$\therefore A^{-1} = \frac{1}{4} \begin{pmatrix} 0 & 0 & -2 \\ 4 & 4 & \frac{1}{3} \\ -1 & 0 & \frac{3}{4} \end{pmatrix} = \begin{pmatrix} 0 & 0 & -\frac{1}{2} \\ 1 & 1 & \frac{1}{12} \\ -\frac{1}{4} & 0 & \frac{3}{16} \end{pmatrix}.$$

36 a)

Calculator:

$$\begin{pmatrix} 2 & 1 & 0 \\ 7 & 1 & 2 \\ -3 & -3 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 7 & -1 & 2 \\ -13 & 2 & -4 \\ -18 & 3 & -5 \end{pmatrix}.$$

By hand:

Find the cofactor matrix:

$$\begin{pmatrix} 7 & -13 & -18 \\ -1 & 2 & 3 \\ 2 & -4 & -5 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 7 & -1 & 2 \\ -13 & 2 & -4 \\ -18 & 3 & -5 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 2 & 1 & 0 \\ 7 & 1 & 2 \\ -3 & -3 & 1 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 \\ -3 & 1 \end{vmatrix} - 1 \begin{vmatrix} 7 & 2 \\ -3 & 1 \end{vmatrix} + 0 \begin{vmatrix} 7 & 1 \\ -3 & -3 \end{vmatrix}$$

$$= 2(1 \times 1 - 2 \times (-3)) - (7 \times 1 - 2 \times (-3)) + 0 = 2 \times 7 - 13 = 1.$$

Then:

$$\begin{pmatrix} 2 & 1 & 0 \\ 7 & 1 & 2 \\ -3 & -3 & 1 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} 7 & -1 & 2 \\ -13 & 2 & -4 \\ -18 & 3 & -5 \end{pmatrix} = \begin{pmatrix} 7 & -1 & 2 \\ -13 & 2 & -4 \\ -18 & 3 & -5 \end{pmatrix}.$$

36 b)

Calculator:

$$\begin{pmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 4 & 3 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{5}{2} & -\frac{1}{2} \\ -2 & -3 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

By hand:

Find the cofactor matrix:

$$\begin{pmatrix} -3 & 4 & 0 \\ -5 & 6 & -2 \\ 1 & -2 & 0 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -3 & -5 & 1 \\ 4 & 6 & -2 \\ 0 & -2 & 0 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 4 & 3 & -1 \end{vmatrix} = 2\begin{vmatrix} 0 & 1 \\ 3 & -1 \end{vmatrix} - 1\begin{vmatrix} 0 & 1 \\ 4 & -1 \end{vmatrix} - 2\begin{vmatrix} 0 & 0 \\ 4 & 3 \end{vmatrix}$$

$$= 2(0 \times (-1) - 1 \times 3) - (0 \times (-1) - 1 \times 4) - 2(0 \times 3 - 0 \times 4)$$

$$= 2 \times (-3) - (-4) - 0 = -2.$$

Then:

$$\begin{pmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 4 & 3 & -1 \end{pmatrix}^{-1} = \frac{1}{-2}\begin{pmatrix} -3 & -5 & 1 \\ 4 & 6 & -2 \\ 0 & -2 & 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & \frac{5}{2} & -\frac{1}{2} \\ -2 & -3 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

36 c)

Calculator:

$$\begin{pmatrix} 4 & 0 & 3 \\ 4 & 3 & 6 \\ 5 & -1 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 5 & -1 & -3 \\ 6 & -1 & -4 \\ -\frac{19}{3} & \frac{4}{3} & 4 \end{pmatrix}.$$

By hand:

Find the cofactor matrix:

$$\begin{pmatrix} 15 & 18 & -19 \\ -3 & -3 & 4 \\ -9 & -12 & 12 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 15 & -3 & -9 \\ 18 & -3 & -12 \\ -19 & 4 & 12 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 4 & 0 & 3 \\ 4 & 3 & 6 \\ 5 & -1 & 3 \end{vmatrix} = 4\begin{vmatrix} 3 & 6 \\ -1 & 3 \end{vmatrix} - 0\begin{vmatrix} 4 & 6 \\ 5 & 3 \end{vmatrix} + 3\begin{vmatrix} 4 & 3 \\ 5 & -1 \end{vmatrix}$$

$$= 4(3 \times 3 - 6 \times (-1)) - 0(4 \times 3 - 6 \times 5) + 3(4 \times (-1) - 3 \times 5)$$
$$= 4 \times 15 - 0 + 3 \times (-19) = 3.$$

Then:
$$\begin{pmatrix} 4 & 0 & 3 \\ 4 & 3 & 6 \\ 5 & -1 & 3 \end{pmatrix}^{-1} = \frac{1}{3}\begin{pmatrix} 15 & -3 & -9 \\ 18 & -3 & -12 \\ -19 & 4 & 12 \end{pmatrix} = \begin{pmatrix} 5 & -1 & -3 \\ 6 & -1 & -4 \\ -\frac{19}{3} & \frac{4}{3} & 4 \end{pmatrix}.$$

37 a)

Calculator:
$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & -3 \\ 1 & 2 & -3 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{3}{5} & \frac{7}{5} & -1 \\ 0 & -1 & 1 \\ \frac{1}{5} & -\frac{1}{5} & 0 \end{pmatrix}.$$

By hand:

Find the cofactor matrix:
$$\begin{pmatrix} 3 & 0 & 1 \\ 7 & -5 & -1 \\ -5 & 5 & 0 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:
$$\begin{pmatrix} 3 & 7 & -5 \\ 0 & -5 & 5 \\ 1 & -1 & 0 \end{pmatrix}.$$

Find the determinant of the original matrix:
$$\begin{vmatrix} 1 & 1 & 2 \\ 1 & 1 & -3 \\ 1 & 2 & -3 \end{vmatrix} = 1\begin{vmatrix} 1 & -3 \\ 2 & -3 \end{vmatrix} - 1\begin{vmatrix} 1 & -3 \\ 1 & -3 \end{vmatrix} + 2\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}$$
$$= 1(1 \times (-3) - (-3) \times 2) - 1(1 \times (-3) - (-3) \times 1) + 2(1 \times 2 - 1 \times 1)$$
$$= 1 \times 3 - 1 \times 0 + 2 \times 1 = 5.$$

Then:
$$A^{-1} = \frac{1}{5}\begin{pmatrix} 3 & 7 & -5 \\ 0 & -5 & 5 \\ 1 & -1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{7}{5} & -1 \\ 0 & -1 & 1 \\ \frac{1}{5} & -\frac{1}{5} & 0 \end{pmatrix}.$$

37 b)

Let P have coordinates (a, b, c).
$$A\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} \Rightarrow A^{-1}A\begin{pmatrix} a \\ b \\ c \end{pmatrix} = A^{-1}\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} \Rightarrow I\begin{pmatrix} a \\ b \\ c \end{pmatrix} = A^{-1}\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$
$$\Rightarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{7}{5} & -1 \\ -1 & 0 & 1 \\ \frac{1}{5} & -\frac{1}{5} & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} \frac{34}{5} \\ -5 \\ -\frac{2}{5} \end{pmatrix}.$$

So P has coordinates $(\frac{34}{5}, -5, -\frac{2}{5})$.

38)

First find B^{-1}:

Calculator:

$$\begin{pmatrix} 6 & -10 & -10 \\ 0 & -3 & -3 \\ 3 & -5 & -\frac{19}{2} \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{6} & -\frac{5}{9} & 0 \\ -\frac{1}{9} & -\frac{1}{3} & \frac{2}{9} \\ \frac{1}{9} & \frac{1}{9} & -\frac{2}{9} \end{pmatrix}.$$

By hand:

Find the cofactor matrix:

$$\begin{pmatrix} \frac{27}{2} & -9 & 9 \\ -45 & -27 & 0 \\ 0 & 18 & -18 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} \frac{27}{2} & -45 & 0 \\ -9 & -27 & 18 \\ 9 & 0 & -18 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 6 & -10 & -10 \\ 0 & -3 & -3 \\ 3 & -5 & -\frac{19}{2} \end{vmatrix} = 6 \begin{vmatrix} -3 & -3 \\ -5 & -\frac{19}{2} \end{vmatrix} + 10 \begin{vmatrix} 0 & -3 \\ 3 & -\frac{19}{2} \end{vmatrix} - 10 \begin{vmatrix} 0 & -3 \\ 3 & -5 \end{vmatrix}$$

$$= 6\left((-3) \times (-\frac{19}{2}) - (-3) \times (-5)\right) + 10(0 \times (-\frac{19}{2}) - (-3) \times 3) - 10(0 \times (-5) - (-3) \times 3)$$

$$= 6 \times \frac{27}{2} + 10 \times 9 - 10 \times 9 = 81.$$

Then:

$$B^{-1} = \frac{1}{81} \begin{pmatrix} \frac{27}{2} & -45 & 0 \\ -9 & -27 & 18 \\ 9 & 0 & -18 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & -\frac{5}{9} & 0 \\ -\frac{1}{9} & -\frac{1}{3} & \frac{2}{9} \\ \frac{1}{9} & 0 & -\frac{2}{9} \end{pmatrix}.$$

$$\begin{pmatrix} -\frac{3}{2} & 5 & 0 \\ 1 & 3 & -2 \\ -1 & 0 & 2 \end{pmatrix} = -9B^{-1}$$

Therefore $A = kB^{-1}$ with $k = -9$.

39 a)

Find the cofactor matrix:

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 1-k \\ 0 & 1-k & k-1 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 1-k \\ 0 & 1-k & k-1 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} k & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = k \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} + 1 \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$$

$$= k(1 \times 0 - 1 \times 1) - 1(1 \times 0 - 1 \times 1) + 1(1 \times 1 - 1 \times 1) = 1 - k.$$

Then:

$$A^{-1} = \frac{1}{1-k}\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 1-k \\ 0 & 1-k & k-1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{1-k} & \frac{1}{1-k} & 0 \\ \frac{1}{1-k} & -\frac{1}{1-k} & 1 \\ 0 & 1 & -1 \end{pmatrix}.$$

39 b)

Find the cofactor matrix:

$$\begin{pmatrix} 4 & -12 & 1 \\ k & 8-2k & -2 \\ -k & 3k & 2 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 4 & k & -k \\ -12 & 8-2k & 3k \\ 1 & -2 & 2 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 2 & 0 & k \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{vmatrix} = 2\begin{vmatrix} 1 & 0 \\ 1 & 4 \end{vmatrix} - 0\begin{vmatrix} 3 & 0 \\ 2 & 4 \end{vmatrix} + k\begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix}$$

$$= 2(1 \times 4 - 0 \times 1) - 0(3 \times 4 - 0 \times 2) + k(3 \times 1 - 1 \times 2) = k + 8.$$

Then:

$$A^{-1} = \frac{1}{k+8}\begin{pmatrix} 4 & k & -k \\ -12 & 8-2k & 3k \\ 1 & -2 & 2 \end{pmatrix} = \begin{pmatrix} \frac{4}{k+8} & \frac{k}{k+8} & -\frac{k}{k+8} \\ -\frac{12}{k+8} & \frac{8-2k}{k+8} & \frac{3k}{k+8} \\ \frac{1}{k+8} & -\frac{2}{k+8} & \frac{2}{k+8} \end{pmatrix}.$$

39 c)

Find the cofactor matrix:

$$\begin{pmatrix} -18 & -3k & 3k \\ -2 & 5k & 3k \\ 14 & k & -9k \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -18 & -2 & 14 \\ -3k & 5k & k \\ 3k & 3k & -9k \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 2k & -1 & 3 \\ k & -5 & 1 \\ k & -2 & 4 \end{vmatrix} = 2k\begin{vmatrix} -5 & 1 \\ -2 & 4 \end{vmatrix} + 1\begin{vmatrix} k & 1 \\ k & 4 \end{vmatrix} + 3\begin{vmatrix} k & -5 \\ k & -2 \end{vmatrix}$$

$$= 2k(-5 \times 4 - 1 \times (-2)) + 1(k \times 4 - 1 \times k) + 3(k \times (-2) - (-5) \times k)$$

$$= 2k \times (-18) + 1 \times 3k + 3 \times 3k = -24k.$$

Then:

$$A^{-1} = \frac{1}{-24k}\begin{pmatrix} -18 & -2 & 14 \\ -3k & 5k & k \\ 3k & 3k & -9k \end{pmatrix} = \begin{pmatrix} \frac{3}{4k} & \frac{1}{12k} & -\frac{7}{12k} \\ \frac{1}{8} & -\frac{5}{24} & -\frac{1}{24} \\ -\frac{1}{8} & -\frac{1}{8} & \frac{3}{8} \end{pmatrix}.$$

39 d)

Find the cofactor matrix:

$$\begin{pmatrix} 3-2k & -2k & 2k+3 \\ -4 & -2 & 0 \\ 2k+3 & 0 & -2k-3 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 3-2k & -4 & 2k+3 \\ -2k & -2 & 0 \\ 2k+3 & 0 & -2k-3 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} -1 & 2 & -1 \\ k & 3 & k \\ -1 & 2 & 1 \end{vmatrix} = -1\begin{vmatrix} 3 & k \\ 2 & 1 \end{vmatrix} - 2\begin{vmatrix} k & k \\ -1 & 1 \end{vmatrix} - 1\begin{vmatrix} k & 3 \\ -1 & 2 \end{vmatrix}$$

$$= -1(3 \times 1 - k \times 2) - 2(k \times 1 - k \times (-1)) - 1(k \times 2 - 3 \times (-1))$$

$$= -1(3-2k) - 2(2k) - 1(2k+3) = -4k - 6.$$

Then:

$$A^{-1} = \frac{1}{-4k-6}\begin{pmatrix} 3-2k & -4 & 2k+3 \\ -2k & -2 & 0 \\ 2k+3 & 0 & -2k-3 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2k-3}{4k+6} & \frac{2}{2k+3} & -\frac{1}{2} \\ \frac{k}{2k+3} & \frac{1}{2k+3} & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}.$$

40 a)

$$\begin{vmatrix} 1 & -1 & k \\ k & 9 & 18 \\ 3 & 0 & 27 \end{vmatrix} = 1\begin{vmatrix} 9 & 18 \\ 0 & 27 \end{vmatrix} + 1\begin{vmatrix} k & 18 \\ 3 & 27 \end{vmatrix} + k\begin{vmatrix} k & 9 \\ 3 & 0 \end{vmatrix}$$

$$= 1(9 \times 27 - 18 \times 0) + 1(k \times 27 - 18 \times 3) + k(k \times 0 - 9 \times 3)$$

$$= 1 \times 243 + 1 \times (27k - 54) + k \times (-27) = 243 + 27k - 54 - 27k = 189.$$

So A is a non-singular matrix for all values of k.

40 b)

Find the cofactor matrix:

$$\begin{pmatrix} 243 & 54-27k & -27 \\ 27 & 27-3k & -3 \\ -9k-18 & k^2-18 & k+9 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 243 & 27 & -9k-18 \\ 54-27k & 27-3k & k^2-18 \\ -27 & -3 & k+9 \end{pmatrix}.$$

Then:

$$A^{-1} = \frac{1}{189}\begin{pmatrix} 243 & 27 & -9k-18 \\ 54-27k & 27-3k & k^2-18 \\ -27 & -3 & k+9 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{9}{7} & \frac{1}{7} & -\frac{k+2}{21} \\ \frac{2-k}{7} & \frac{9-k}{63} & \frac{k^2-18}{189} \\ -\frac{1}{7} & -\frac{1}{63} & \frac{k+9}{189} \end{pmatrix}.$$

41)

Find the cofactor matrix:

$$\begin{pmatrix} 5-5k & 5k-5 & k^2+k-2 \\ 18-2k & 5k-27 & k^2-4k-9 \\ 16 & 3k-25 & -2k-10 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 5-5k & 18-2k & 16 \\ 5k-5 & 5k-27 & 3k-25 \\ k^2+k-2 & k^2-4k-9 & -2k-10 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 5-k & 4 & -2 \\ k & -2 & 5 \\ -1 & k+1 & -5 \end{vmatrix} = (5-k)\begin{vmatrix} -2 & 5 \\ k+1 & -5 \end{vmatrix} - 4\begin{vmatrix} k & 5 \\ -1 & -5 \end{vmatrix} - 2\begin{vmatrix} k & -2 \\ -1 & k+1 \end{vmatrix}$$

$$= (5-k)(-2\times(-5) - 5\times(k+1)) - 4(k\times(-5) - 5\times(-1)) - 2(k\times(k+1) - (-2)\times(-1))$$

$$= (5-k)(5-5k) - 4(5-5k) - 2(k^2+k-2)$$

$$= 5(5-k)(1-k) - 20(1-k) - 2(k+2)(k-1)$$

$$= 5(k-5)(k-1) + 20(k-1) - 2(k+2)(k-1)$$

$$= (k-1)(5(k-5) + 20 - 2(k+2)) = (k-1)(3k-9) = 3(k-1)(k-3).$$

Then:

$$A^{-1} = \frac{1}{3(k-1)(k-3)} \begin{pmatrix} 5-5k & 18-2k & 16 \\ 5k-5 & 5k-27 & 3k-25 \\ k^2+k-2 & k^2-4k-9 & -2k-10 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{5}{3(k-3)} & -\frac{2(k-9)}{3(k-1)(k-3)} & \frac{16}{3(k-1)(k-3)} \\ \frac{5}{3(k-3)} & \frac{5k-27}{3(k-1)(k-3)} & \frac{3k-25}{3(k-1)(k-3)} \\ \frac{k+2}{3(k-3)} & \frac{k^2-4k-9}{3(k-1)(k-3)} & -\frac{2(k+5)}{3(k-1)(k-3)} \end{pmatrix}.$$

A is non-singular if $k \neq 1$ or $k \neq 3$.

42)

Find the cofactor matrix:

$$\begin{pmatrix} -k-5 & 1 & -1 \\ 3k+7 & -k^2-5k-5 & -2k-1 \\ -3k-3 & k^2+4k+3 & 3k+3 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -k-5 & 3k+7 & -3k-3 \\ 1 & -k^2-5k-5 & k^2+4k+3 \\ -1 & -2k-1 & 3k+3 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} -2-k & -3 & 1 \\ -1 & -3 & k+2 \\ -1 & -2 & k+3 \end{vmatrix} = (-2-k)\begin{vmatrix} -3 & k+2 \\ -2 & k+3 \end{vmatrix} + 3\begin{vmatrix} -1 & k+2 \\ -1 & k+3 \end{vmatrix} + \begin{vmatrix} -1 & -3 \\ -1 & -2 \end{vmatrix}$$

$$= (-2-k)((-3)(k+3)-(k+2)(-2))+3((-1)(k+3)-(k+2)(-1))+((-1)(-2)-(-3)(-1))$$
$$= (-2-k)(-k-5)+3(-1)+1(-1) = k^2+7k+6 = (k+1)(k+6).$$

Then:

$$A^{-1} = \frac{1}{(k+1)(k+6)}\begin{pmatrix} -k-5 & 3k+7 & -3k-3 \\ 1 & -k^2-5k-5 & k^2+4k+3 \\ -1 & -2k-1 & 3k+3 \end{pmatrix}$$
$$= \begin{pmatrix} -\frac{k+5}{(k+1)(k+6)} & \frac{3k+7}{(k+1)(k+6)} & -\frac{3}{k+6} \\ \frac{1}{(k+1)(k+6)} & -\frac{k^2+5k+5}{(k+1)(k+6)} & \frac{k+3}{k+6} \\ -\frac{1}{(k+1)(k+6)} & -\frac{2k+1}{(k+1)(k+6)} & \frac{3}{k+6} \end{pmatrix}, \quad k \neq -1,\ k \neq -6.$$

43)

Find the cofactor matrix:

$$\begin{pmatrix} 2k-9 & 0 & 0 \\ k^2-24 & 3 & -k \\ 3k-16 & 2 & -3 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 2k-9 & k^2-24 & 3k-16 \\ 0 & 3 & 2 \\ 0 & -k & -3 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 1 & 8 & k \\ 0 & -3 & -2 \\ 0 & k & 3 \end{vmatrix} = 1\begin{vmatrix} -3 & -2 \\ k & 3 \end{vmatrix} - 8\begin{vmatrix} 0 & -2 \\ 0 & 3 \end{vmatrix} + k\begin{vmatrix} 0 & -3 \\ 0 & k \end{vmatrix}$$

$$= 1((-3)\times 3-(-2)\times k)-8(0\times 3-(-2)\times 0)+k(0\times k-(-3)\times 0) = 2k-9.$$

Then:

$$A^{-1} = \frac{1}{2k-9}\begin{pmatrix} 2k-9 & k^2-24 & 3k-16 \\ 0 & 3 & 2 \\ 0 & -k & -3 \end{pmatrix} = \begin{pmatrix} 1 & \frac{k^2-24}{2k-9} & \frac{3k-16}{2k-9} \\ 0 & \frac{3}{2k-9} & \frac{2}{2k-9} \\ 0 & -\frac{k}{2k-9} & -\frac{3}{2k-9} \end{pmatrix}.$$

$$\frac{k^2-24}{2k-9} = 8 \Rightarrow k^2-24 = 16k-72 \Rightarrow k^2-16k+48 = 0 \Rightarrow k = 4 \text{ or } k = 12$$

$$\frac{3k-16}{2k-9} = k \Rightarrow 3k-16 = 2k^2-9k \Rightarrow 2k^2-12k+16 = 0 \Rightarrow k = 2 \text{ or } k = 4$$

$$\frac{3}{2k-9} = -3 \Rightarrow 3 = -6k+27 \Rightarrow 6k = 24 \Rightarrow k = 4$$

$$\frac{2}{2k-9} = -2 \Rightarrow 2 = -4k+18 \Rightarrow 4k = 16 \Rightarrow k = 4$$

$$-\frac{k}{2k-9} = k \Rightarrow -k = 2k^2-9k \Rightarrow 2k^2-8k = 0 \Rightarrow k = 0 \text{ or } k = 4$$

$$-\frac{3}{2k-9} = 3 \Rightarrow -3 = 6k-27 \Rightarrow 6k = 24 \Rightarrow k = 4.$$

$$\therefore k = 4.$$

44 a)

$$\begin{pmatrix}2 & 3\\1 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}5\\4\end{pmatrix}\Rightarrow\begin{pmatrix}2 & 3\\1 & 1\end{pmatrix}^{-1}\begin{pmatrix}2 & 3\\1 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}2 & 3\\1 & 1\end{pmatrix}^{-1}\begin{pmatrix}5\\4\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}2 & 3\\1 & 1\end{pmatrix}^{-1}\begin{pmatrix}5\\4\end{pmatrix}\Rightarrow\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}2 & 3\\1 & 1\end{pmatrix}^{-1}\begin{pmatrix}5\\4\end{pmatrix}$$

$$\begin{vmatrix}2 & 3\\1 & 1\end{vmatrix}=2\times1-3\times1=-1$$

$$\begin{pmatrix}2 & 3\\1 & 1\end{pmatrix}^{-1}=\frac{1}{-1}\begin{pmatrix}1 & -3\\-1 & 2\end{pmatrix}=\begin{pmatrix}-1 & 3\\1 & -2\end{pmatrix}$$

$$\therefore\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}-1 & 3\\1 & -2\end{pmatrix}\begin{pmatrix}5\\4\end{pmatrix}=\begin{pmatrix}7\\-3\end{pmatrix}.$$

So $x=7$ and $y=-3$.

44 b)

$$\begin{pmatrix}3 & 1\\1 & 2\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}-3\\9\end{pmatrix}\Rightarrow\begin{pmatrix}3 & 1\\1 & 2\end{pmatrix}^{-1}\begin{pmatrix}3 & 1\\1 & 2\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}3 & 1\\1 & 2\end{pmatrix}^{-1}\begin{pmatrix}-3\\9\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}3 & 1\\1 & 2\end{pmatrix}^{-1}\begin{pmatrix}-3\\9\end{pmatrix}\Rightarrow\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}3 & 1\\1 & 2\end{pmatrix}^{-1}\begin{pmatrix}-3\\9\end{pmatrix}$$

$$\begin{vmatrix}3 & 1\\1 & 2\end{vmatrix}=3\times2-1\times1=5$$

$$\begin{pmatrix}3 & 1\\1 & 2\end{pmatrix}^{-1}=\frac{1}{5}\begin{pmatrix}2 & -1\\-1 & 3\end{pmatrix}=\begin{pmatrix}\frac{2}{5} & -\frac{1}{5}\\-\frac{1}{5} & \frac{3}{5}\end{pmatrix}$$

$$\therefore\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\frac{2}{5} & -\frac{1}{5}\\-\frac{1}{5} & \frac{3}{5}\end{pmatrix}\begin{pmatrix}-3\\9\end{pmatrix}=\begin{pmatrix}-3\\6\end{pmatrix}.$$

So $x=-3$ and $y=6$.

44 c)

$$\begin{pmatrix}4 & -1\\8 & -3\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}15\\1\end{pmatrix}\Rightarrow\begin{pmatrix}4 & -1\\8 & -3\end{pmatrix}^{-1}\begin{pmatrix}4 & -1\\8 & -3\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}4 & -1\\8 & -3\end{pmatrix}^{-1}\begin{pmatrix}15\\1\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}4 & -1\\8 & -3\end{pmatrix}^{-1}\begin{pmatrix}15\\1\end{pmatrix}\Rightarrow\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}4 & -1\\8 & -3\end{pmatrix}^{-1}\begin{pmatrix}15\\1\end{pmatrix}$$

$$\begin{vmatrix}4 & -1\\8 & -3\end{vmatrix}=4\times(-3)-(-1)\times8=-4$$

$$\begin{pmatrix}4 & -1\\8 & -3\end{pmatrix}^{-1}=\frac{1}{-4}\begin{pmatrix}-3 & 1\\-8 & 4\end{pmatrix}=\begin{pmatrix}\frac{3}{4} & -\frac{1}{4}\\2 & -1\end{pmatrix}$$

$$\therefore\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\frac{3}{4} & -\frac{1}{4}\\2 & -1\end{pmatrix}\begin{pmatrix}15\\1\end{pmatrix}=\begin{pmatrix}11\\29\end{pmatrix}.$$

So $x=11$ and $y=29$.

44 d)

$$\begin{pmatrix} -6 & 2 \\ 9 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -7 \\ -10 \end{pmatrix} \Rightarrow \begin{pmatrix} -6 & 2 \\ 9 & -4 \end{pmatrix}^{-1}\begin{pmatrix} -6 & 2 \\ 9 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -6 & 2 \\ 9 & -4 \end{pmatrix}^{-1}\begin{pmatrix} -7 \\ -10 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -6 & 2 \\ 9 & -4 \end{pmatrix}^{-1}\begin{pmatrix} -7 \\ -10 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -6 & 2 \\ 9 & -4 \end{pmatrix}^{-1}\begin{pmatrix} -7 \\ -10 \end{pmatrix}$$

$$\begin{vmatrix} -6 & 2 \\ 9 & -4 \end{vmatrix} = (-6) \times (-4) - 2 \times 9 = 6$$

$$\begin{pmatrix} -6 & 2 \\ 9 & -4 \end{pmatrix}^{-1} = \frac{1}{6}\begin{pmatrix} -4 & -2 \\ -9 & -6 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} & -\frac{1}{3} \\ -\frac{3}{2} & -1 \end{pmatrix}$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} & -\frac{1}{3} \\ -\frac{3}{2} & -1 \end{pmatrix}\begin{pmatrix} -7 \\ -10 \end{pmatrix} = \begin{pmatrix} 8 \\ \frac{41}{2} \end{pmatrix}.$$

So $x = 8$ and $y = \frac{41}{2}$.

45)

$$\begin{vmatrix} 2 & 9k \\ k & 18 \end{vmatrix} = 2 \times 18 - 9k \times k = 36 - 9k^2$$

No unique solution when $\begin{vmatrix} 2 & 9k \\ k & 18 \end{vmatrix} = 0 \Rightarrow 36 - 9k^2 = 0 \Rightarrow k = \pm 2.$

When $k = 2$:

$$2x + 18y = 6 \qquad 2x + 18y = 6$$

So the two lines are coincident when $k = 2$.

When $k = -2$:

$$2x - 18y = 6 \qquad -2x + 18y = 6 \Rightarrow 2x - 18y = -6$$

So the two lines are parallel when $k = -2$.

46)

$$\begin{vmatrix} 8k & 48 \\ 3 & 2k \end{vmatrix} = 8k \times 2k - 48 \times 3 = 16k^2 - 144$$

No unique solution when $\begin{vmatrix} 8k & 48 \\ 3 & 2k \end{vmatrix} = 0 \Rightarrow 16k^2 - 144 = 0 \Rightarrow k = \pm 3.$

When $k = 3$:

$$24x + 48y = 5 \Rightarrow 3x + 6y = \frac{5}{8} \qquad 3x + 6y = 3$$

So the two lines are parallel when $k = 3$.

When $k = -3$:

$$-24x + 48y = 5 \Rightarrow 3x - 6y = -\frac{5}{8} \qquad 3x - 6y = 3$$

So the two lines are parallel when $k = -3$.

47 a)

The coefficients of x, y and z match the elements of matrix A when we divide through by 10:

$$0.2x + 0y - 0.3z = 4$$

$$0x - 0.2y + 0.3z = -6$$

$$0.1x - 0.2y - 0.1z = 9.$$

Given that we know that A^{-1} exists, the three simultaneous equations must have a unique solution (as $\det A \neq 0$).

47 b)

$$A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \\ 9 \end{pmatrix} \Rightarrow A^{-1}A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} 4 \\ -6 \\ 9 \end{pmatrix} \Rightarrow I\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} 4 \\ -6 \\ 9 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 & 6 & -6 \\ 3 & 1 & -6 \\ 2 & 4 & -4 \end{pmatrix}\begin{pmatrix} 4 \\ -6 \\ 9 \end{pmatrix} = \begin{pmatrix} -58 \\ -48 \\ -52 \end{pmatrix}.$$

So $x = -58$, $y = -48$ and $z = -52$.

48 a)

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & -1 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & -1 & 4 \end{pmatrix}^{-1}\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & -1 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & -1 & 4 \end{pmatrix}^{-1}\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$\Rightarrow I\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & -1 & 4 \end{pmatrix}^{-1}\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & -1 & 4 \end{pmatrix}^{-1}\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

Calculator:

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & -1 & 4 \end{pmatrix}^{-1} = \begin{pmatrix} -\frac{1}{2} & \frac{3}{5} & \frac{1}{10} \\ \frac{1}{2} & \frac{1}{5} & -\frac{3}{10} \\ \frac{1}{2} & -\frac{2}{5} & \frac{1}{10} \end{pmatrix}.$$

By hand:

Find the cofactor matrix:

$$\begin{pmatrix} 5 & -5 & -5 \\ -6 & -2 & 4 \\ -1 & 3 & -1 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 5 & -6 & -1 \\ -5 & -2 & 3 \\ -5 & 4 & -1 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & -1 & 4 \end{vmatrix} = 1\begin{vmatrix} 1 & 1 \\ -1 & 4 \end{vmatrix} - 1\begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} + 2\begin{vmatrix} 2 & 1 \\ 3 & -1 \end{vmatrix}$$

$= 1(1 \times 4 - 1 \times (-1)) - 1(2 \times 4 - 1 \times 3) + 2(2 \times (-1) - 1 \times 3) = 1 \times 5 - 1 \times 5 + 2 \times (-5) = -10.$

Then:

$$A^{-1} = \frac{1}{-10}\begin{pmatrix} 5 & -6 & -1 \\ -5 & -2 & 3 \\ -5 & 4 & -1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{3}{5} & \frac{1}{10} \\ \frac{1}{2} & \frac{1}{5} & -\frac{3}{10} \\ \frac{1}{2} & -\frac{2}{5} & \frac{1}{10} \end{pmatrix}.$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{3}{5} & \frac{1}{10} \\ \frac{1}{2} & \frac{1}{5} & -\frac{3}{10} \\ \frac{1}{2} & -\frac{2}{5} & \frac{1}{10} \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

So $x = \frac{3}{2}$, $y = \frac{1}{2}$ and $z = -\frac{1}{2}$.

48 b)

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 0 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 0 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 0 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 0 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 5 \\ 8 \\ 3 \end{pmatrix}$$

$$\Rightarrow I \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 0 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 5 \\ 8 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 0 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 5 \\ 8 \\ 3 \end{pmatrix}$$

Calculator:

$$A^{-1} = \begin{pmatrix} \frac{5}{8} & \frac{5}{8} & -\frac{1}{8} \\ \frac{3}{8} & -\frac{5}{8} & \frac{1}{8} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

By hand: Find the cofactor matrix:

$$\begin{pmatrix} -5 & -3 & 2 \\ -5 & 5 & 2 \\ 1 & -1 & -2 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -5 & -5 & 1 \\ -3 & 5 & -1 \\ 2 & 2 & -2 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 0 & 5 \end{vmatrix} = 1\begin{vmatrix} -1 & 1 \\ 0 & 5 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 2 & 5 \end{vmatrix} + 0\begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix}$$

$$= 1((-1) \times 5 - 1 \times 0) - 1(1 \times 5 - 1 \times 2) + 0 = -5 - 3 = -8.$$

Then:

$$A^{-1} = \frac{1}{-8}\begin{pmatrix} -5 & -5 & 1 \\ -3 & 5 & -1 \\ 2 & 2 & -2 \end{pmatrix} = \begin{pmatrix} \frac{5}{8} & \frac{5}{8} & -\frac{1}{8} \\ \frac{3}{8} & -\frac{5}{8} & \frac{1}{8} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

$$\therdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{5}{8} & \frac{5}{8} & -\frac{1}{8} \\ \frac{3}{8} & -\frac{5}{8} & \frac{1}{8} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 5 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} \frac{31}{4} \\ -\frac{11}{4} \\ -\frac{5}{2} \end{pmatrix}.$$

So $x = \frac{31}{4}$, $y = -\frac{11}{4}$ and $z = -\frac{5}{2}$.

48 c)

$$A\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-7\\3\\-2\end{pmatrix}$$

$$\Rightarrow A^{-1}A\begin{pmatrix}x\\y\\z\end{pmatrix}=A^{-1}\begin{pmatrix}-7\\3\\-2\end{pmatrix}$$

$$\Rightarrow I\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}0&1&-3\\-1&0&4\\-1&-1&1\end{pmatrix}^{-1}\begin{pmatrix}-7\\3\\-2\end{pmatrix}\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}0&1&-3\\-1&0&4\\-1&-1&1\end{pmatrix}^{-1}\begin{pmatrix}-7\\3\\-2\end{pmatrix}$$

Calculator:

$$A^{-1}=\begin{pmatrix}-\frac{2}{3}&-\frac{1}{3}&-\frac{2}{3}\\\frac{1}{2}&\frac{1}{2}&-\frac{1}{2}\\-\frac{1}{6}&\frac{1}{6}&-\frac{1}{6}\end{pmatrix}.$$

By hand: Find the cofactor matrix:

$$\begin{pmatrix}4&-3&1\\2&-3&-1\\4&3&1\end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix}4&2&4\\-3&-3&3\\1&-1&1\end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix}0&1&-3\\-1&0&4\\-1&-1&1\end{vmatrix}=0\begin{vmatrix}0&4\\-1&1\end{vmatrix}-1\begin{vmatrix}-1&4\\-1&1\end{vmatrix}-3\begin{vmatrix}-1&0\\-1&-1\end{vmatrix}$$

$=0(0\times1-4\times(-1))-1((-1)\times1-4\times(-1))-3((-1)\times(-1)-0\times(-1))=0-1\times3-3\times1=-6.$

Then:

$$A^{-1}=\frac{1}{-6}\begin{pmatrix}4&2&4\\-3&-3&3\\1&-1&1\end{pmatrix}=\begin{pmatrix}-\frac{2}{3}&-\frac{1}{3}&-\frac{2}{3}\\\frac{1}{2}&\frac{1}{2}&-\frac{1}{2}\\-\frac{1}{6}&\frac{1}{6}&-\frac{1}{6}\end{pmatrix}.$$

$$\therefore\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-\frac{2}{3}&-\frac{1}{3}&-\frac{2}{3}\\\frac{1}{2}&\frac{1}{2}&-\frac{1}{2}\\-\frac{1}{6}&\frac{1}{6}&-\frac{1}{6}\end{pmatrix}\begin{pmatrix}-7\\3\\-2\end{pmatrix}=\begin{pmatrix}5\\-1\\2\end{pmatrix}.$$

So $x=5$, $y=-1$ and $z=2$.

49)

$$\begin{pmatrix}2&1&-1&4\\2&-3&3&5\\6&-2&-9&-7\\1&-8&3&2\end{pmatrix}\begin{pmatrix}a\\b\\c\\d\end{pmatrix}=\begin{pmatrix}8\\14\\11\\6\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 1 & -1 & 4 \\ 2 & -3 & 3 & 5 \\ 6 & -2 & -9 & -7 \\ 1 & -8 & 3 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 2 & 1 & -1 & 4 \\ 2 & -3 & 3 & 5 \\ 6 & -2 & -9 & -7 \\ 1 & -8 & 3 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 2 & 1 & -1 & 4 \\ 2 & -3 & 3 & 5 \\ 6 & -2 & -9 & -7 \\ 1 & -8 & 3 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix}$$

$$\Rightarrow I \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 2 & 1 & -1 & 4 \\ 2 & -3 & 3 & 5 \\ 6 & -2 & -9 & -7 \\ 1 & -8 & 3 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 2 & 1 & -1 & 4 \\ 2 & -3 & 3 & 5 \\ 6 & -2 & -9 & -7 \\ 1 & -8 & 3 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{2} & \frac{143}{578} & \frac{1}{2} & -\frac{91}{578} \\ \frac{97}{289} & -\frac{15}{578} & -\frac{88}{289} & -\frac{43}{578} \\ 0 & \frac{60}{289} & -\frac{1}{2} & \frac{145}{578} \\ \frac{43}{289} & \frac{22}{289} & \frac{64}{289} & \frac{5}{578} \end{pmatrix} \begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 5 \\ 2 \end{pmatrix}.$$

So $a = 3$, $b = -2$, $c = 5$ and $d = 2$.

50 a)

Find the cofactor matrix:

$$\begin{pmatrix} 0 & 1-k & 0 \\ -k & k^2-1 & 1 \\ 1 & 1-k & -1 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 0 & -k & 1 \\ 1-k & k^2-1 & 1-k \\ 0 & 1 & -1 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} k & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & k \end{vmatrix} = k \begin{vmatrix} 0 & 1 \\ 0 & k \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 1 & k \end{vmatrix} + 1 \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}$$

$$= k(0 \times k - 1 \times 0) - 1(1 \times k - 1 \times 1) + 1(1 \times 0 - 0 \times 1) = 1 - k.$$

Then:

$$A^{-1} = \frac{1}{1-k} \begin{pmatrix} 0 & -k & 1 \\ 1-k & k^2-1 & 1-k \\ 0 & 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -\frac{k}{1-k} & \frac{1}{1-k} \\ 1 & -k-1 & 1 \\ 0 & \frac{1}{1-k} & -\frac{1}{1-k} \end{pmatrix}.$$

50 b)

When $k = 3$, $A^{-1} = \begin{pmatrix} 0 & \frac{3}{2} & -\frac{1}{2} \\ 1 & -4 & 1 \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$.

$$A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -6 \\ 18 \\ -12 \end{pmatrix} \Rightarrow A^{-1}A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} -6 \\ 18 \\ -12 \end{pmatrix} \Rightarrow I\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} -6 \\ 18 \\ -12 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & \frac{3}{2} & -\frac{1}{2} \\ 1 & -4 & 1 \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} -6 \\ 18 \\ -12 \end{pmatrix} = \begin{pmatrix} 33 \\ -90 \\ -15 \end{pmatrix}.$$

So $x = 33$, $y = -90$ and $z = -15$.

51 a)

Find the cofactor matrix:

$$\begin{pmatrix} 5 & 2k-3 & 2-3k \\ 0 & -3k-15 & 2k+10 \\ -5 & 13 & -17 \end{pmatrix}.$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 5 & 0 & -5 \\ 2k-3 & -3k-15 & 13 \\ 2-3k & 2k+10 & -17 \end{pmatrix}.$$

Find the determinant of the original matrix:

$$\begin{vmatrix} -5 & 2 & 3 \\ 1 & 3 & 2 \\ k & 2 & 3 \end{vmatrix} = -5\begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} - 2\begin{vmatrix} 1 & 2 \\ k & 3 \end{vmatrix} + 3\begin{vmatrix} 1 & 3 \\ k & 2 \end{vmatrix}$$

$= -5(3\times 3 - 2\times 2) - 2(1\times 3 - 2\times k) + 3(1\times 2 - 3\times k) = -5\times 5 - 2(3-2k) + 3(2-3k) = -5k - 25.$

Then:

$$A^{-1} = \frac{1}{-5k-25}\begin{pmatrix} 5 & 0 & -5 \\ 2k-3 & -3k-15 & 13 \\ 2-3k & 2k+10 & -17 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{k+5} & 0 & \frac{1}{k+5} \\ -\frac{2k-3}{5k+25} & \frac{3}{5} & -\frac{13}{5k+25} \\ -\frac{2-3k}{5k+25} & -\frac{2}{5} & \frac{17}{5k+25} \end{pmatrix}.$$

51 b)

When $k = 5$, $A^{-1} = \begin{pmatrix} -\frac{1}{10} & 0 & \frac{1}{10} \\ -\frac{7}{50} & \frac{3}{5} & -\frac{13}{50} \\ \frac{13}{50} & -\frac{2}{5} & \frac{17}{50} \end{pmatrix}$.

$$A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \\ 9 \end{pmatrix} \Rightarrow A^{-1}A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} 2 \\ -5 \\ 9 \end{pmatrix} \Rightarrow I\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} 2 \\ -5 \\ 9 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{1}{10} & 0 & \frac{1}{10} \\ -\frac{7}{50} & \frac{3}{5} & -\frac{13}{50} \\ \frac{13}{50} & -\frac{2}{5} & \frac{17}{50} \end{pmatrix}\begin{pmatrix} 2 \\ -5 \\ 9 \end{pmatrix} = \begin{pmatrix} \frac{7}{10} \\ -\frac{281}{50} \\ \frac{279}{50} \end{pmatrix}.$$

So $x = \frac{7}{10}$, $y = -\frac{281}{50}$ and $z = \frac{279}{50}$.

52 a)

$$C^{-1}CAB^{-1}B = (C^{-1}C)A(B^{-1}B) = IAI = A.$$

52 b)

$$B(AB)^{-1} = B(B^{-1}A^{-1}) = (BB^{-1})A^{-1} = IA^{-1} = A^{-1}.$$

52 c)

$$(CA^{-1})^{-1}C = (AC^{-1})C = A(C^{-1}C) = AI = A.$$

52 d)

$$C(ABC)^{-1}A = C(C^{-1}B^{-1}A^{-1})A = (CC^{-1})B^{-1}(A^{-1}A) = IB^{-1}I = B^{-1}.$$

52 e)

$$(B^{-1}A)^{-1}(C^{-1}B)^{-1} = (A^{-1}B)(B^{-1}C) = A^{-1}(BB^{-1})C = A^{-1}IC = A^{-1}C.$$

52 f)

$$(AB^{-1}C)^{-1}(C^{-1}BA^{-1})^{-1} = (C^{-1}BA^{-1})(AB^{-1}C) = C^{-1}B(A^{-1}A)B^{-1}C$$
$$= C^{-1}BIB^{-1}C = C^{-1}(BB^{-1})C = C^{-1}IC = C^{-1}C = I.$$

52 g)

$$(ACBA^{-1})^{-1}(AB^{-1}C^{-1}A^{-1})^{-1} = (AB^{-1}C^{-1}A^{-1})(ACBA^{-1})$$
$$= AB^{-1}C^{-1}(A^{-1}A)CBA^{-1} = AB^{-1}C^{-1}ICBA^{-1} = AB^{-1}(C^{-1}C)BA^{-1}$$
$$= AB^{-1}IBA^{-1} = A(B^{-1}B)A^{-1} = AIA^{-1} = AA^{-1} = I.$$

53)

$$AB = C \Rightarrow A^{-1}AB = A^{-1}C \Rightarrow IB = A^{-1}C \Rightarrow B = A^{-1}C.$$

54)

$$ABC = A \Rightarrow A^{-1}ABC = A^{-1}A \Rightarrow IBC = I \Rightarrow BC = I$$
$$\Rightarrow BCC^{-1} = IC^{-1} \Rightarrow BI = C^{-1} \Rightarrow B = C^{-1}.$$

55 a)

Matrix multiplication is not generally commutative, so equating $B^{-1}ABA^{-1}$ to $B^{-1}BAA^{-1}$ is incorrect (i.e. $AB \neq BA$).

55 b)

$$(B^{-1}A)^{-1} = A^{-1}B.$$

55 c)

$$\begin{gathered}(B^{-1}A)(B^{-1}A)^{-1} = I \Rightarrow BB^{-1}A(B^{-1}A)^{-1} = BI \\ \Rightarrow IA(B^{-1}A)^{-1} = B \Rightarrow A(B^{-1}A)^{-1} = B \\ \Rightarrow A^{-1}A(B^{-1}A)^{-1} = A^{-1}B \Rightarrow I(B^{-1}A)^{-1} = A^{-1}B \Rightarrow (B^{-1}A)^{-1} = A^{-1}B.\end{gathered}$$

56)

$$\begin{gathered}B = A^{-1}CA \Rightarrow AB = AA^{-1}CA \Rightarrow AB = ICA \Rightarrow AB = CA \\ \Rightarrow ABA^{-1} = CAA^{-1} \Rightarrow ABA^{-1} = CI \Rightarrow C = ABA^{-1}.\end{gathered}$$

57)

$$\begin{gathered}ABA = I \Rightarrow A^{-1}ABA = A^{-1}I \Rightarrow IBA = A^{-1} \Rightarrow BA = A^{-1} \\ \Rightarrow BAA^{-1} = A^{-1}A^{-1} \Rightarrow BI = A^{-1}A^{-1} \Rightarrow B = A^{-1}A^{-1}.\end{gathered}$$

58)

$$\begin{gathered}(AB)(AB)^{-1} = I \Rightarrow A^{-1}AB(AB)^{-1} = A^{-1}I \Rightarrow IB(AB)^{-1} = A^{-1} \Rightarrow B(AB)^{-1} = A^{-1} \\ \Rightarrow B^{-1}B(AB)^{-1} = B^{-1}A^{-1} \Rightarrow I(AB)^{-1} = B^{-1}A^{-1} \Rightarrow (AB)^{-1} = B^{-1}A^{-1}.\end{gathered}$$

59)

$$\begin{gathered}(ABC)(ABC)^{-1} = I \Rightarrow A^{-1}ABC(ABC)^{-1} = A^{-1}I \Rightarrow IBC(ABC)^{-1} = A^{-1} \\ \Rightarrow BC(ABC)^{-1} = A^{-1} \Rightarrow B^{-1}BC(ABC)^{-1} = B^{-1}A^{-1} \Rightarrow IC(ABC)^{-1} = B^{-1}A^{-1} \\ \Rightarrow C(ABC)^{-1} = B^{-1}A^{-1} \Rightarrow C^{-1}C(ABC)^{-1} = C^{-1}B^{-1}A^{-1} \\ \Rightarrow I(ABC)^{-1} = C^{-1}B^{-1}A^{-1} \Rightarrow (ABC)^{-1} = C^{-1}B^{-1}A^{-1}.\end{gathered}$$

60)

$$\begin{gathered}(A^{-1}B^{-1}C^{-1})(A^{-1}B^{-1}C^{-1})^{-1} = I \Rightarrow AA^{-1}B^{-1}C^{-1}(A^{-1}B^{-1}C^{-1})^{-1} = AI \\ \Rightarrow IB^{-1}C^{-1}(A^{-1}B^{-1}C^{-1})^{-1} = A \Rightarrow B^{-1}C^{-1}(A^{-1}B^{-1}C^{-1})^{-1} = A \\ \Rightarrow BB^{-1}C^{-1}(A^{-1}B^{-1}C^{-1})^{-1} = BA \Rightarrow IC^{-1}(A^{-1}B^{-1}C^{-1})^{-1} = BA \\ \Rightarrow C^{-1}(A^{-1}B^{-1}C^{-1})^{-1} = BA \Rightarrow CC^{-1}(A^{-1}B^{-1}C^{-1})^{-1} = CBA \\ \Rightarrow I(A^{-1}B^{-1}C^{-1})^{-1} = CBA \Rightarrow (A^{-1}B^{-1}C^{-1})^{-1} = CBA.\end{gathered}$$

61)

$$\det(AB) = \det A \times \det B.$$

If $B = A^{-1}$, then

$$\det(AA^{-1}) = \det A \times \det(A^{-1}) \Rightarrow \det(I) = \det A \times \det(A^{-1})$$

$$\Rightarrow 1 = \det A \times \det(A^{-1}) \Rightarrow \det(A^{-1}) = \frac{1}{\det A}.$$

62)

$$(AA^{-1})^T = (A^{-1})^T A^T \Rightarrow I^T = (A^{-1})^T A^T \Rightarrow I = (A^{-1})^T A^T$$
$$\Rightarrow I(A^T)^{-1} = (A^{-1})^T A^T (A^T)^{-1}$$
$$\Rightarrow (A^T)^{-1} = (A^{-1})^T I \Rightarrow (A^T)^{-1} = (A^{-1})^T.$$

63 a)

$$AB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \times \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad-bc} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$
$$= \frac{1}{ad-bc} \begin{pmatrix} ad-bc & -ab+ab \\ cd-cd & ad-bc \end{pmatrix} = \frac{1}{ad-bc} \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix}$$
$$= \begin{pmatrix} \frac{ad-bc}{ad-bc} & 0 \\ 0 & \frac{ad-bc}{ad-bc} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

63 b)

$$BA = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad-bc} \begin{pmatrix} ad-bc & bd-bd \\ -ac+ac & ad-bc \end{pmatrix}$$
$$= \frac{1}{ad-bc} \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} = \begin{pmatrix} \frac{ad-bc}{ad-bc} & 0 \\ 0 & \frac{ad-bc}{ad-bc} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

64)

Let $A^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$.

$$AA^{-1} = I \Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} ap+br & aq+bs \\ cp+dr & cq+ds \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$ap + br = 1 \quad (1) \qquad aq + bs = 0 \quad (2)$$
$$cp + dr = 0 \quad (3) \qquad cq + ds = 1 \quad (4)$$

$$(1): \quad p = \frac{1-br}{a}$$

$$(3): \quad p = -\frac{dr}{c}$$

$$\therefore \frac{1-br}{a} = -\frac{dr}{c} \Rightarrow c - bcr = -adr \Rightarrow c = adr - bcr \Rightarrow (ad-bc)r = -c \Rightarrow r = -\frac{c}{ad-bc}$$

$$\therefore p = -\frac{d}{c} \times -\frac{c}{ad-bc} = \frac{d}{ad-bc}.$$

$$(2): \quad q = -\frac{bs}{a}$$

$$(4): \quad q = \frac{1-ds}{c}$$

$$\therefore \frac{1-ds}{c} = -\frac{bs}{a} \Rightarrow a - ads = -bcs \Rightarrow a = ads - bcs \Rightarrow a = (ad-bc)s \Rightarrow s = \frac{a}{ad-bc}$$

$$\therefore q = -\frac{b}{a} \times \frac{a}{ad-bc} = -\frac{b}{ad-bc}.$$

So:

$$A^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$